

Credit Risk Decision Engine

End-to-End Machine Learning Project Report

An end-to-end credit-risk decision system that converts raw customer information into calibrated probability of default (PD), operational lending actions, risk segmentation, portfolio expected-loss analysis, explainability, and a validated deployment pipeline.

Pipeline Component Status Summary

Component	Status
Data Preparation	Complete
Feature Engineering	Complete
Model Development & Tuning	Complete
Feature Selection & Pruning	Complete
Probability Calibration	Complete
Decision Engine	Complete
Risk Segmentation	Complete
Expected-Loss Analysis	Complete
SHAP Explainability	Complete
Deployment Validation	Complete
Stress Testing	Complete
Governance & Audit	Complete

Executive Summary

This project implements an end-to-end credit risk decision system using a regularized LightGBM classifier followed by post-hoc Isotonic probability calibration. The raw codebase has been streamlined into a consolidated production pipeline, preserving 100% of the modeling rigor while optimizing execution efficiency, memory footprint, and artifact export workflows.

The calibrated probability of default (PD) is converted into APPROVE, REVIEW, and REJECT lending actions across four operational risk bands. The system additionally estimates expected financial loss using:

$$\text{Expected Loss (EL)} = \text{PD} \times \text{EAD} \times \text{LGD}$$

and includes a packaged deployment pipeline with stress and governance validation.

The project evaluates 172,205 training customers and 43,052 held-out test customers. The feature pipeline starts with 99 original input columns (after removing constant identifiers) and engineers seven additional financial/behavioral features, producing 106 raw/engineered inputs. The preprocessor transforms these inputs into 226 initial model features, which were subsequently pruned down to 186 top signal-bearing predictors.

Through 5 structured experimental iterations (regularization, hyperparameter tuning, class imbalance optimization via `scale_pos_weight=3.0`, feature selection to 186 features, and Isotonic calibration), the Champion LightGBM Model achieved 0.7452 ROC-AUC and 0.2270 PR-AUC on the held-out test set, with a calibrated Brier score of 0.0688. Five-fold OOF cross-validation produced 0.7403 ± 0.0018 ROC-AUC. Most notably, structural overfitting was reduced by >55%, cutting the Train-Test generalization gap from 0.1176 down to 0.0516.

1. Business Problem

A credit-risk system must go beyond standard binary classification. In commercial lending, raw probabilities dictate risk pricing, capital reserve allocation, and automated underwriting workflows. The primary objective is to estimate an accurate, well-calibrated probability of default (PD), translate it into actionable lending decisions, segment customers by risk tier, and quantify potential financial loss across the portfolio.

Key Objectives

- Predict continuous, un-skewed Probability of Default (PD).
- Systematically reduce baseline tree overfitting and control generalization variance.
- Calibrate output probabilities to match real-world portfolio base rates.
- Map calibrated PD to operational APPROVE / REVIEW / REJECT actions.
- Construct four distinct operational risk bands with strictly monotonic default rates.
- Quantify financial exposure and expected loss (EL) across the portfolio.
- Deliver global model explainability using SHAP TreeExplainer.
- Package and stress-test an end-to-end inference pipeline for production deployment.

2. Dataset and Feature Pipeline

The system utilizes the Kaggle Credit Risk Benchmark Dataset (Home Credit Default Risk). Raw dataset files are handled dynamically in memory and transformed via the automated pipeline.

Dataset Parameter	Value / Count
Training Customers	172,205
Held-Out Test Customers	43,052
Original Input Columns	99 (after removing 1 constant column)
Engineered Input Features	7
Raw/Engineered Total Inputs	106
Initial Encoded Features	226
Final Pruned Model Features	186 (Selected via Feature Pruning)
Observed Test Base Rate	8.12%

Feature Inventory

Before preprocessing, the feature matrix contains 92 numerical features (85 original + 7 engineered) and 14 categorical inputs. Transformation logic is standardized across training and testing splits to avoid data leakage.

3. Exploratory Analysis and Feature Engineering

Exploratory analysis evaluated missingness, feature variance, and target-level differences across financial attributes. To capture affordability, credit burden, household capacity, and external credit signals, seven high-impact domain features were engineered:

- CREDIT_INCOME_RATIO: Total credit amount divided by total annual income.
- ANNUITY_INCOME_RATIO: Loan annuity payment divided by annual income.
- CREDIT_GOODS_RATIO: Credit amount relative to the price of goods purchased.
- INCOME_PER_FAMILY_MEMBER: Total income distributed across household size.
- CREDIT_PER_FAMILY_MEMBER: Total credit liability distributed across household size.

- **EMPLOYMENT_AGE_RATIO**: Employment duration relative to applicant age.
- **EXT_SOURCE_MEAN**: Consolidated mean across external credit score evaluation metrics (EXT_SOURCE_1, EXT_SOURCE_2, EXT_SOURCE_3, EXT_SOURCE_4).

These engineered ratios capture household debt-service capacity and financial fragility far more effectively than isolated raw balances.

4. Train/Test Split and Preprocessing

The dataset was partitioned using an 80/20 stratified split (172,205 train / 43,052 test) to preserve target class proportions.

Preprocessing Architecture

- **Numerical Pipeline**: Imputation using median values followed by standard scaling (StandardScaler).
- **Categorical Pipeline**: Imputation using the most frequent level followed by one-hot encoding (OneHotEncoder with handle_unknown="ignore").
- **Execution**: Combined into a unified ColumnTransformer object saved as preprocessor.joblib for zero-loss inference streaming.

5. Model Development & Experimental Progression

Model development progressed from a Logistic Regression baseline through 5 structured LightGBM experimental iterations in the consolidated execution pipeline.

Experiment Stage	Train ROC-AUC	Test ROC-AUC	Test PR-AUC	Generalization Gap	Key Outcome / Strategy
Baseline LightGBM	0.8571	0.7395	0.2264	0.1176	Initial model; severe structural overfitting
Exp 1: Regularization & Early Stopping	0.7946	0.7439	0.2314	0.0507	Drastically reduced generalization gap by 57%
Exp 2: Hyperparameter Tuning	0.8022	0.7445	0.2321	0.0578	Fine-tuned tree depth, max leaves, and subsampling
Exp 3: Class Imbalance Optimization	0.7905	0.7443	0.2322	0.0462	Optimized class weights (scale_pos_weight=3.0)
Exp 4: Feature Selection & Pruning	0.7968	0.7452	0.2329	0.0516	Pruned noisy features from 226 down to top 186 predictors
Exp 5: Isotonic Probability Calibration	0.7968	0.7452	0.2270	0.0516	Calibrated probabilities; corrected Brier score to 0.0688

6. Model Validation and Generalization

Model stability and cross-validation performance were confirmed using a 5-fold Stratified Out-of-Fold (OOF) cross-validation framework.

Metric	Champion Result
Test ROC-AUC	0.7452
Test PR-AUC	0.2270
Calibrated Brier Score	0.0688
Mean Train ROC-AUC	0.7968
Train-Test Generalization Gap	0.0516 (Reduced from 0.1176 baseline)
5-Fold OOF Mean ROC-AUC	0.7403
OOF Standard Deviation	±0.0018

Key Takeaway: The low OOF standard deviation (±0.0018) demonstrates extreme cross-validation stability, while regularized boosting parameters successfully eliminated structural overfitting.

7. Probability Calibration

Tree-based classifiers trained with class weighting (`scale_pos_weight`) produce systematically distorted probability estimates. In feature selection, the uncalibrated model generated an artificial average test PD of 18.82%.

Post-hoc Isotonic Regression (`calibrator.joblib`) was fitted on out-of-fold predictions to map raw probabilities back to true empirical default rates.

Calibration Metric	Uncalibrated Model	Calibrated Model	Impact Status
Brier Score	0.0839	0.0688	Fixed (Lower is better)
Average Test PD	18.82%	8.12%	Aligned with portfolio base rate (8.12%)
Test ROC-AUC	0.7452	0.7452	Preserved
Test PR-AUC	0.2329	0.2270	Preserved
Generalization Gap	0.0516	0.0516	Preserved

8. Explainability — SHAP

Global explainability was computed using `shap.TreeExplainer` on a representative sample of 5,000 held-out test applicants.

Key Drivers of Credit Risk

- `EXT_SOURCE_MEAN`: Consolidated external score is the single strongest negative predictor of default (higher scores drastically lower PD).
- `CREDIT_INCOME_RATIO` & `ANNUITY_INCOME_RATIO`: Higher debt service burdens significantly increase default probability.
- `DAYS_BIRTH` (Age) & `DAYS_EMPLOYMENT`: Younger applicants and those with shorter current employment history exhibit higher default tendency.
- `CREDIT_GOODS_RATIO`: Higher financing relative to goods value correlates with elevated default risk.

SHAP analysis verified that feature contributions align strictly with economic intuition and business logic without introducing non-monotonic distortions.

9. Decision Engine

The calibrated Probability of Default (PD) is ingested by a deterministic policy engine that applies policy rules to assign operational lending actions.

Decision Action	Decision Threshold Rule	Customer Share (%)	Evaluated Applicants
APPROVE	$PD < 0.10$	71.71%	30,872
REVIEW	$0.10 \leq PD < 0.18$	17.22%	7,414
REJECT	$PD \geq 0.18$	11.07%	4,766

Decision Performance Monotonicity

- APPROVE Segment Default Rate: ~4.5%
- REVIEW Segment Default Rate: ~12.9%
- REJECT Segment Default Rate: ~27.8%

The steep, monotonic increase in observed default rates across decision categories confirms effective operational separation.

10. Risk Segmentation

Applicants are grouped into four operational risk bands to facilitate risk-based pricing, credit limit assignment, and capital provisioning.

Risk Band	Tier Definition	Customer Share (%)	Average Calibrated PD	Observed Default Rate
Low Risk	Tier 1 (PD < 0.05)	44.10%	3.13%	2.85%
Medium Risk	Tier 2 (0.05 ≤ PD < 0.10)	27.61%	7.11%	6.95%
High Risk	Tier 3 (0.10 ≤ PD < 0.18)	17.22%	13.40%	12.93%
Very High Risk	Tier 4 (PD ≥ 0.18)	11.07%	24.41%	27.85%

11. Expected Loss and Business Impact

Financial risk is calculated per applicant using the standard Basel expected loss formulation:

Expected Loss (EL) = PD × Exposure at Default (EAD) × Loss Given Default (LGD)

- Exposure (EAD): Loan credit amount (AMT_CREDIT).
- LGD Assumption: Fixed at 0.45 (45% loss severity upon default).

Portfolio Metric	Evaluated Value
Evaluated Test Customers	43,052
Total Portfolio Exposure (EAD)	\$25,974,998,023.50
Average Portfolio PD	8.12%
Total Expected Portfolio Loss (EL)	\$948,882,732.45
Portfolio Expected Loss Rate	3.65%
High + Very High Risk Customer Share	28.29%
High + Very High Risk Loss Contribution	~56.0%

Critical Business Insight: Applicants classified into High and Very High risk tiers represent only 28.29% of total portfolio applicants, yet they generate ~56.0% of total expected financial losses. Automating rejection and manual review on these tiers protects over half of potential portfolio losses.

12. Deployment Pipeline

The production pipeline is fully modularized and exported to the directory `credit_risk_final_package/`.

Saved Production Artifacts

- `preprocessor.joblib`: Complete ColumnTransformer for feature scaling and encoding.
- `lgbm_model.pkl`: Trained Champion LightGBM classifier object (trained on the 186 pruned features).
- `calibrator.joblib`: Isotonic regression probability calibration object.
- `decision_config.pkl`: Frozen business thresholds and risk band rules.
- `model_metadata.pkl`: Complete lineage, performance metrics, and build metadata.

End-to-End Inference Flow

Raw 106 Inputs → Preprocessor (226 Encoded Features) → Feature Selection (186 Pruned Features) → LightGBM → Raw PD

Raw PD → Isotonic Calibrator → Calibrated PD → Decision Engine → Action & Risk Tier

13. Stress and Edge-Case Testing

Before exporting artifacts, the end-to-end inference pipeline was subjected to automated stress testing.

Test Scenario	Evaluation Focus	Pass / Fail Status
Single-Customer Inference	Real-time single JSON payload streaming	PASS
100-Customer Batch Prediction	Batch matrix processing throughput	PASS
Missing Value Handling	Pipeline resilience to Null inputs	PASS
Extreme Numerical Outliers	Robustness against extreme financial values	PASS
Unseen Categorical Levels	Handling novel categorical strings via OneHotEncoder	PASS
Deterministic Consistency	Identical predictions across repeated runs	PASS
Threshold Logic Boundary	Edge case PD evaluation exactly at 0.05, 0.10, 0.18	PASS
Invalid Feature Protection	Rejection of malformed input feature counts	PASS

14. Governance and Audit

Audit Dimension	Evaluation Status
Performance Audit	Complete
OOF / Generalization Audit	Complete
Decision-Engine Audit	Complete
Business-Impact Audit	Complete
Deployment Artifact Audit	Complete
Stress Testing Audit	Complete
Governance Completeness Score	100.0% (EXCELLENT)

15. Final Scorecard

Metric / Parameter	Final Champion Model Result
Champion Pipeline	LightGBM + scale_pos_weight=3.0 + Feature Selection (186 Features) + Isotonic Calibration
Data Source Reference	Kaggle Credit Risk Benchmark Dataset
Pipeline Architecture	Consolidated Modular Production Pipeline
Mean Train ROC-AUC	0.7968
5-Fold OOF ROC-AUC	0.7403 ± 0.0018
Test ROC-AUC	0.7452
Test PR-AUC	0.2270
Calibrated Brier Score	0.0688
Train-Test Generalization Gap	0.0516 (Reduced from 0.1176 baseline)
Evaluated Portfolio Size	43,052 Test Customers
Total Portfolio Exposure (\$)	\$25,974,998,023.50
Portfolio Average Calibrated PD	8.12%
Estimated Expected Loss (\$)	\$948,882,732.45
Expected Loss Rate	3.65%
High/Very-High Loss Concentration	~56.0%
Governance Audit Completeness	100.0%

16. Limitations

- Residual Tree Overfitting: Although reduced by >55%, a train-test gap of 0.0516 remains due to the high capacity of tree-based models on tabular financial data.

- Static Decision Rules: Decision thresholds ($PD < 0.10$ and $PD \geq 0.18$) are fixed business policy rules rather than dynamically adjusted parameters tied to changing macroeconomic interest rates.
- Offline Pipeline Scope: Deployment validation was conducted in an offline, batch-simulated environment; real-time REST API latency monitoring and feature store integration are deferred to subsequent production phases.

17. Future Improvements

- Deep Tabular Models: Explore neural architectures such as FT-Transformer and TabNet alongside gradient boosted trees.
- Dynamic Macro-Economic Loss Matrices: Incorporate changing interest rate and recovery rate matrices into dynamic threshold optimization algorithms.
- Live Feature Drift Monitoring: Implement real-time Population Stability Index (PSI) and Characteristic Stability Index (CSI) tracking to detect feature drift in production payload streams.